

3.1 Introduction, Design Issues and Duties of Physical Layer

The **physical layer** is the lowest layer in the OSI and TCP/IP networking models. Its main responsibility is to move raw bits, represented as 0s and 1s, from one device to the next over a physical transmission medium. The medium may be copper cable, optical fiber, or wireless space. At this layer, the network does not yet understand webpages, files, account numbers, email messages, or database records. It only handles signals that represent bits.

Higher layers decide what information must be sent, but the physical layer decides how the information physically travels from one point to another.

The physical layer coordinates the functions required to carry a bit stream over a physical medium. It deals with mechanical and electrical specifications of interfaces and transmission media and defines the procedures that physical devices must perform for transmission. In simple classroom language, the physical layer answers questions such as: What connector is used? What type of cable or wireless frequency is used? How fast are bits sent? How are 0s and 1s converted into electrical, optical, or radio signals?

3.1.1 Duties of the Physical Layer

A network communication system can work properly only when both the sender and receiver follow the same physical rules. The physical layer establishes these rules. Its duties are not only technical but also practical because wrong cable type, wrong speed setting, poor signal quality, or improper synchronization can make a network unstable even when all software settings are correct.

Table 1: Major Duties of the Physical Layer

Duty	Meaning	Practical Example
Physical characteristics of interfaces and media	Defines the type of cable, connector, pin layout, wireless frequency, voltage level, and mechanical characteristics used to connect devices.	A bank branch switch uses RJ-45 ports for copper Ethernet and SFP ports for fiber uplinks.
Representation of bits	Defines how 0s and 1s are converted into signals such as voltage levels, light pulses, or radio waves.	A network card converts digital data from a core banking terminal into electrical pulses on an Ethernet cable.
Data rate	Defines how many bits are sent per second. It is commonly measured in bps, kbps, Mbps, or Gbps.	A 1 Gbps LAN port can transmit up to one billion bits per second under ideal conditions.
Synchronization of bits	Ensure that sender and receiver clocks match so that the receiver reads the bits at the correct time.	If clocks are poorly synchronized, a receiver may misread 1010 as a different bit sequence.
Line configuration	Defines whether the link is point-to-point or multipoint.	A dedicated fiber link between data center and disaster recovery site is point-to-point. A shared Wi-Fi channel is multipoint.
Physical topology	Defines how devices are physically connected: bus, star, ring, mesh, or hybrid.	Most corporate LANs use a star topology with switches at the center.
Transmission mode	Defines the direction of data flow: simplex, half-duplex, or full-duplex.	Modern switched Ethernet is commonly full-duplex, allowing simultaneous sending and receiving.

3.1.2 Design Issues of the Physical Layer

Physical layer design involves several engineering decisions. These decisions directly affect network cost, speed, distance, reliability, and security. For example, a small office may use copper twisted-pair cable because it is cheap and easy to install, while a bank data center may use optical fiber because it supports higher bandwidth, lower attenuation, and stronger resistance to electromagnetic noise.

- **Medium selection:** Copper, fiber, or wireless must be selected based on distance, bandwidth, cost, physical environment, and security requirements.
- **Signal encoding:** The system must decide how bits will be represented as signal changes. Bad encoding can waste bandwidth or make synchronization difficult.
- **Distance limitation:** Every medium has a maximum practical distance before the signal becomes too weak or distorted.
- **Noise protection:** The design must reduce electromagnetic interference, crosstalk, and signal corruption.
- **Connector and hardware compatibility:** Cable, connector, transceiver, switch port, and network interface card must match.
- **Scalability:** The physical network should support future speed upgrades, for example from 1 Gbps to 10 Gbps.
- **Reliability and redundancy:** Critical networks often use redundant links so that one cable failure does not stop the service.
- **Physical security:** Data center cables, fiber routes, wireless access points, and network cabinets must be physically protected.

Common misunderstanding: Many beginners think networking problems are always caused by IP addresses, software, or server configuration. In reality, many network problems start at the physical layer: loose cables, damaged fiber, wrong transceiver, incorrect duplex setting, weak Wi-Fi signal, or electromagnetic interference.

Key Takeaways

- The physical layer is responsible for moving individual bits from one node to the next.
- It defines media, interfaces, signals, data rate, synchronization, topology, and transmission mode.
- It does not interpret the meaning of data; it only transmits bit patterns physically.
- Physical layer faults can cause slow speed, packet loss, link failure, and unstable communication.
- Good physical layer design is essential for reliable banking, cloud, campus, and data center networks.

3.2 Transmission Media: Guided and Unguided Media

3.2.1 Meaning of Transmission Media

A **transmission medium** is the path through which data signals travel from sender to receiver. It is the physical or wireless channel used for communication. Transmission media are commonly classified into **guided media** and **unguided media**. Guided media provides a physical path such as copper or fiber. Unguided media transmit electromagnetic waves through air or free space.

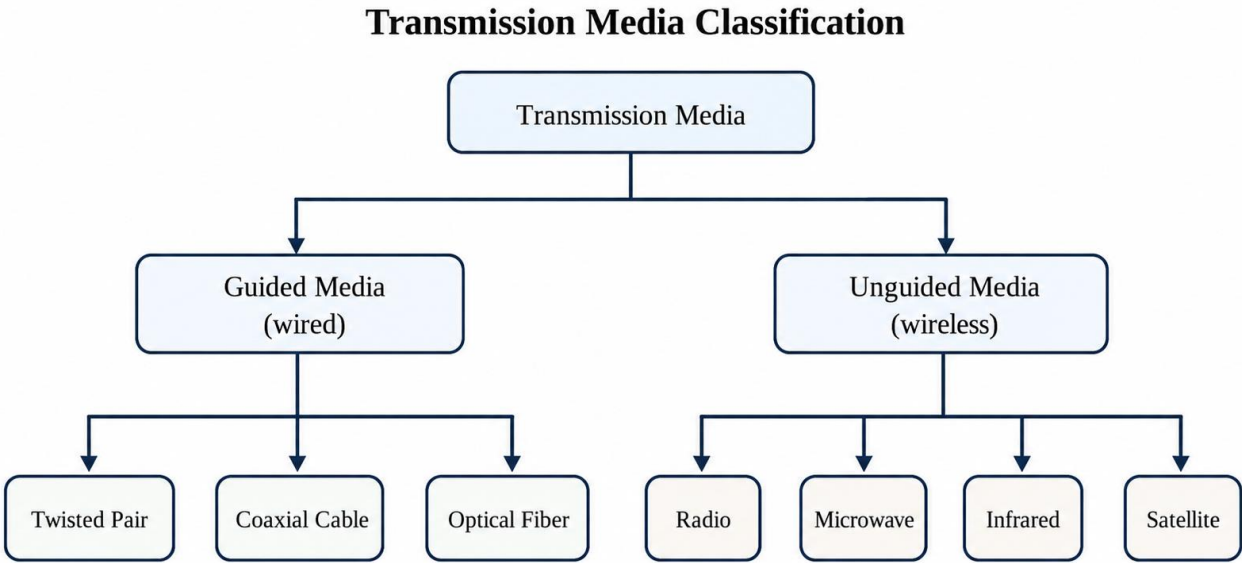


Figure 3: Classification of Transmission Media

Table 2: Guided Media vs Unguided Media

Basis	Guided Media	Unguided Media
Physical path	Uses a solid physical conductor or conduit such as copper or glass fiber.	Uses air or free space; no physical conductor is required between sender and receiver.
Examples	Twisted-pair cable, coaxial cable, optical fiber.	Radio Waves, microwaves, infrared Waves, satellite links.
Direction control	Signal is guided along the cable or fiber.	Signal spreads through space; direction depends on antenna design.
Security	Usually more physically controllable, but tapping is still possible.	More exposed to interception if not properly encrypted.
Mobility	Limited mobility because devices must be physically connected.	Supports mobility and flexible deployment.
Typical use	LAN cabling, data center interconnects, backbone links.	Wi-Fi, Bluetooth, mobile networks, satellite communication.

3.2.2 Twisted-Pair Cable

Twisted-pair cable consists of two insulated copper wires twisted together. The twisting is not decorative; it reduces electromagnetic interference and crosstalk. When two wires are twisted, external noise tends to affect both wires in a similar way, allowing the receiver to reduce the effect of that noise. Twisted-pair cable is widely used in telephone systems, Ethernet LANs, office networks, and campus networks.

Twisted-Pair Cable Concept

Two insulated copper wires are twisted together to reduce interference and crosstalk.

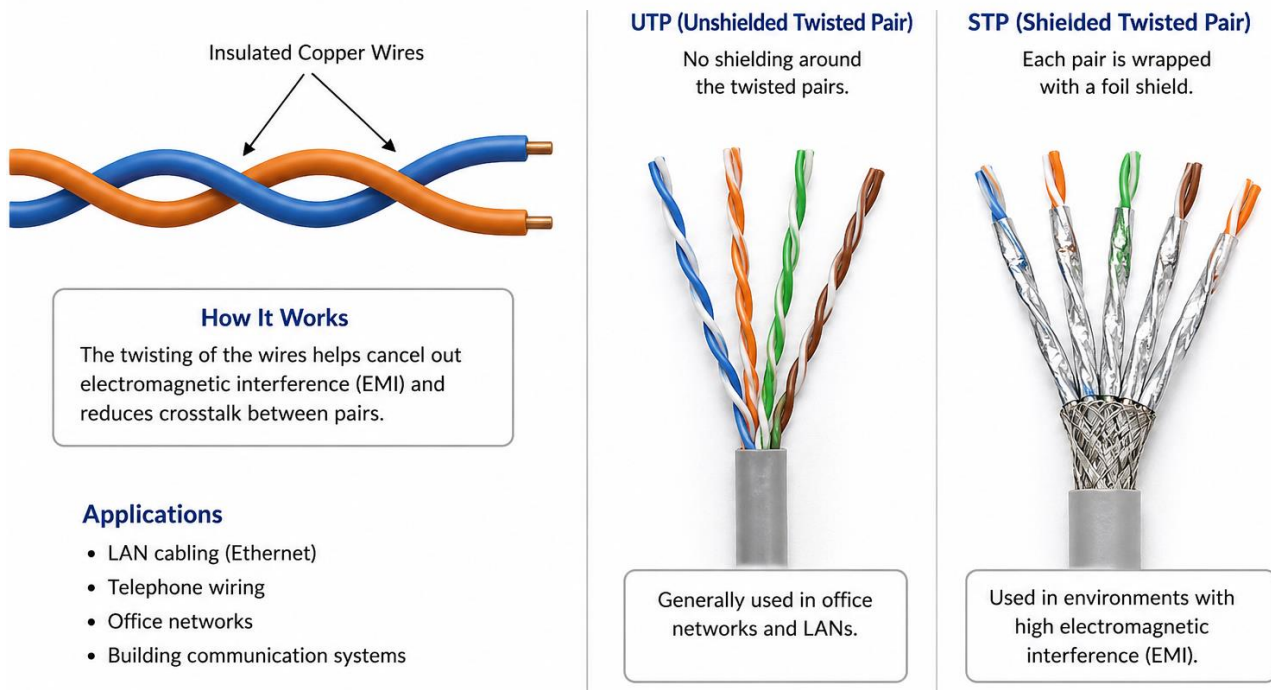


Figure 4: Twisted-Pair Cable Concept

There are two common forms of twisted-pair cable: **unshielded twisted pair (UTP)** and **shielded twisted pair (STP)**. UTP is cheaper, easier to install, and common in office LANs. STP includes additional shielding and is useful in electrically noisy environments, but it is more expensive and harder to install properly.

- **Purpose:** To carry electrical signals for voice and data communication over short to moderate distances.
- **How it works:** Bits are represented by electrical signal changes traveling along copper conductors.
- **Key characteristics:** Low cost, flexible, easy installation, widely supported by Ethernet equipment, but more affected by noise and distance limitations than fiber.
- **Limitations:** Signal attenuation increases with distance; electromagnetic interference can affect performance; typical Ethernet copper runs are limited to about 100 meters for many common LAN standards.

Example: In a bank branch, employee computers, printers, biometric devices, and IP phones may be connected to access switches using UTP Ethernet cables. The cables usually terminate at patch panels and wall outlets using RJ-45 connectors.

3.2.3 Coaxial Cable

Coaxial cable has a central conductor surrounded by insulation, a metallic shield, and an outer jacket. A coaxial cable has two conductors that share the same axis means the inner copper conductor is placed at the center, and the outer metallic shield surrounds it in a circular form, both aligned along the same central line (same axis).

Because of its shielding and structure, coaxial cable can carry higher-frequency signals than ordinary twisted-pair cable and is more resistant to interference. Historically, coaxial cable was used in traditional Ethernet LANs and is still widely used in cable television and broadband access networks.

COAXIAL CABLE

A coaxial cable has two conductors that share the same axis.

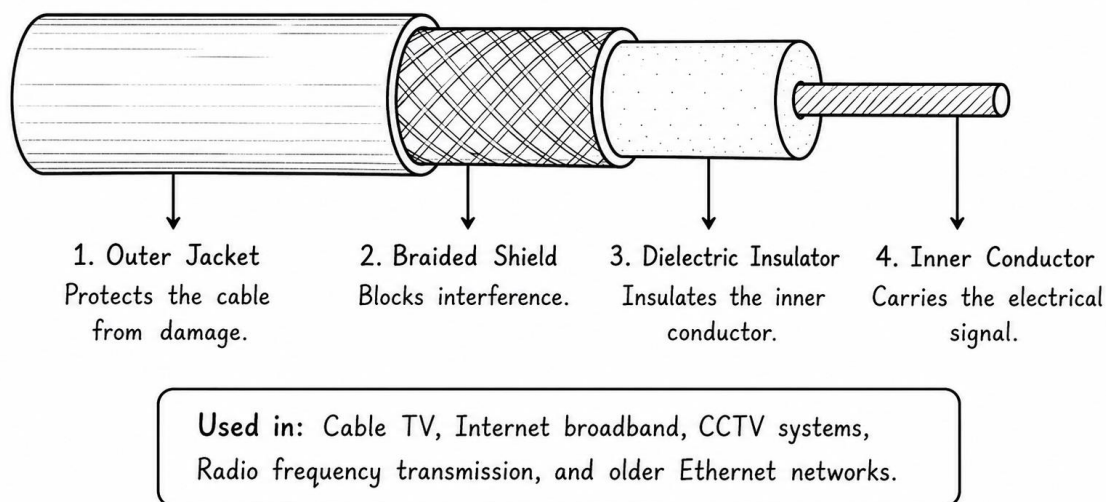


Figure 5: Coaxial Cable Structure

Purpose

The main purpose of coaxial cable is to transmit electrical signals with better protection from external noise and interference than ordinary twisted-pair cable.

It is especially useful where the signal must travel over a longer distance or where the environment has more electrical disturbance.

How It Works

Coaxial cable works by sending electrical signals through the **central copper conductor**.

The signal is surrounded by an insulating layer called the **dielectric insulator**. Around this insulator, there is a **metallic shield** or braided conductor. This shield reduces the effect of external noise and prevents the signal from leaking outside the cable.

The outer jacket protects the cable from physical damage, moisture, and environmental effects.

The basic signal flow is:

1. The sender generates an electrical signal.
2. The central conductor carries the signal forward.
3. The dielectric insulator separates the inner conductor from the outer shield.
4. The metallic shield blocks electromagnetic interference.
5. The receiver receives the signal with less noise and distortion.

Key Characteristics

- Coaxial cable has a **central conductor** and an **outer conductor** sharing the same axis.
- It provides better shielding than ordinary twisted-pair cable.
- It is suitable for high-frequency signal transmission.
- It is more resistant to **electromagnetic interference (EMI)**.
- It is physically stronger than twisted-pair cable.
- It is thicker, heavier, and less flexible than twisted-pair cable.
- It was used in older Ethernet LANs.
- It is still widely used in cable TV and broadband internet systems.

In older LAN environments, Ethernet standards such as **10BASE5** and **10BASE2** used coaxial cable to connect computers in a shared network.

3.2.4 Optical Fiber

Optical fiber cable is a guided transmission medium that carries data in the form of light signals instead of electrical signals.

It is mainly used for high-speed, long-distance communication, such as internet backbone, data centers, banking networks, telecom networks, and FTTH internet.

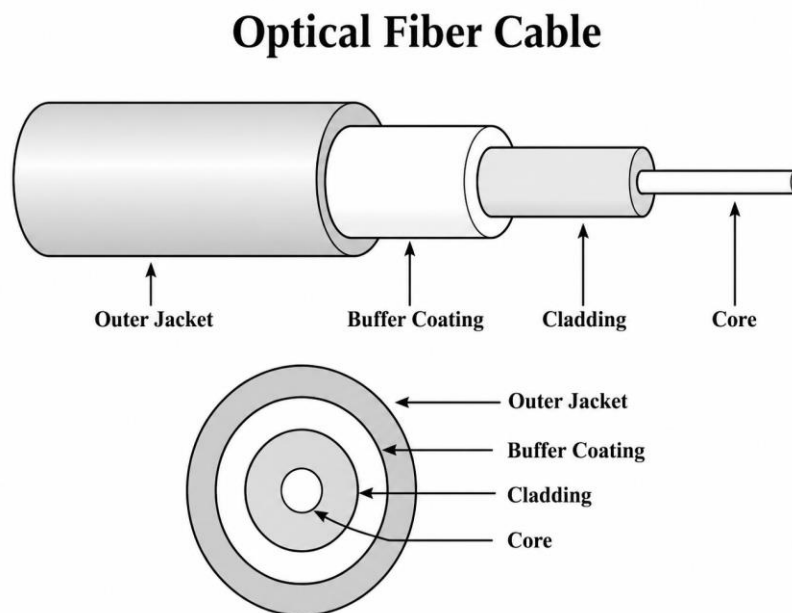


Figure 6: Optical Fiber and Total Internal Reflection

Structure of Optical Fiber Cable

The diagram shows four main parts:

Outer Jacket

The **outer jacket** is the outside protective covering of the fiber cable.

Function:

- Protects the cable from physical damage
- Protects from moisture, dust, and external pressure
- Makes the cable stronger for installation

It is like the plastic cover of an electric wire.

Buffer Coating

The **buffer coating** is a protective layer around the inner fiber.

Function:

- Protects the delicate glass fiber from bending and breaking
- Gives mechanical strength
- Help prevent scratches and cracks

It acts like a cushion around the glass fiber.

Cladding

The **cladding** is the glass or plastic layer surrounding the core.

Function:

- Keeps the light inside the core
- Helps light reflect into the core
- Supports a process called **total internal reflection**

The cladding has a slightly lower refractive index than the core. Because of this difference, light does not easily escape from the core.

Core

The **core** is the central part of the optical fiber.

Function:

- Carries the actual light signal
- Data travels through this part as pulses of light

The core is usually made of very pure glass or plastic. The core is the main road where the light signal travels.

- **Purpose:** To carry high-speed data over longer distances using light instead of electrical current.
- **How it works:** A transmitter converts bits into light pulses; the pulses travel through the core by reflection; a receiver converts light pulses back into electrical or digital signals.
- **Key characteristics:** High bandwidth, low attenuation, lightweight cable, immunity to electromagnetic interference, suitable for backbone networks and data centers.
- **Limitations:** Higher installation cost, more delicate termination, specialized equipment required, and skilled handling needed for splicing and testing.

Example: A bank may connect its primary data center and disaster recovery site using redundant fiber links. Fiber is preferred because core banking, ATM switching, card systems, digital banking, and replication traffic require high capacity and reliability.

Common misunderstanding: fiber is always faster simply because it is fiber. In reality, speed depends on the complete system: fiber type, transceiver, switch port, wavelength, distance, and network standard.

Table 3: Comparison of Guided Transmission Media

Medium	Strengths	Limitations	Common Applications
Twisted Pair	Low cost, easy installation, common RJ-45 Ethernet support.	Shorter distance, more affected by EMI and crosstalk.	Office LANs, IP phones, branch networks, campus wiring.
Coaxial Cable	Better shielding than twisted pair, supports higher frequency.	Less flexible, less common in modern LANs.	Cable TV, cable Internet, older Ethernet.
Optical Fiber	Very high bandwidth, long distance, low attenuation, EMI immunity.	Higher cost, specialized installation/testing.	Backbone links, data centers, ISP networks, DR links.

3.2.5 Radio Waves

Radio waves are electromagnetic waves used for wireless communication. They are generally **omnidirectional**, meaning they can spread in many directions from the transmitting antenna. This makes them useful for broadcasting and mobile communication where the receiver may not remain fixed in one position.

Radio waves are generally considered to cover the lower part of the wireless electromagnetic spectrum, approximately from **3 kHz to 1 GHz**.

Lower-frequency radio waves spread broadly and are useful when the receiver may be moving or may not be directly aligned with the transmitter.

Radio waves are used to support wireless communication in areas where installing physical cables is difficult, expensive, or impractical, and where users or devices need mobility. In radio wave communication, a transmitter uses an antenna to radiate electromagnetic waves through free space. These waves travel through the air and are captured by the receiver antenna, which converts them back into electrical signals for processing. Radio waves can cover wide geographical areas, support mobile users, and penetrate buildings depending on their frequency. However, because they travel through open space, they may be affected by interference, signal weakening, and security exposure. Radio waves are widely used in radio broadcasting, television broadcasting, mobile communication, walkie-talkies, wireless sensors, and IoT systems.

Example:

FM radio, AM radio, television broadcasting, walkie-talkies, and mobile communication use radio-frequency signals.

3.2.6 Microwave

Microwaves are high-frequency electromagnetic waves used for wireless communication. They are usually **directional**, meaning the signal is focused on a particular receiver rather than spreading equally in all directions. Microwaves generally range from about **1 GHz to 300 GHz**. Microwaves have higher frequency and shorter wavelength than radio waves. Because of this, they are usually more

directional and commonly require **line-of-sight communication**. This means the transmitting and receiving antennas should face each other without major obstacles between them.

Because of this directional nature, microwave links are often used between towers, buildings, branches, or network sites for point-to-point communication.

Microwave communication is used to provide high-capacity wireless links in situations where laying fiber or copper cable is expensive, difficult, or time-consuming. It works by using directional antennas that focus microwave energy on the receiver, instead of spreading the signal widely in all directions. This focused transmission reduces unnecessary signal loss and improves link quality. Microwave communication is suitable for point-to-point links, such as communication between towers, buildings, branches, or network sites. However, it commonly requires a clear line-of-sight between the transmitting and receiving antennas. Its performance can be affected by obstacles, weather conditions, antenna misalignment, distance, and regulatory frequency planning.

Example:

A bank may use a microwave link as a backup connection for a remote branch where leased fiber is unavailable or takes a long time to repair.

Note:

Wi-Fi and Bluetooth commonly use microwave-frequency radio signals, especially in the 2.4 GHz and 5 GHz bands.

3.2.7 Infrared

Infrared communication uses infrared light for short-range wireless communication. It is usually used indoors and typically requires devices to be close to each other. Infrared signals do not penetrate walls effectively, which limits their range but also reduces interference between rooms. Infrared waves generally range from about **300 GHz to 400 THz**.

Infrared communication is used to support short-range wireless communication between nearby devices. It works by transmitting light pulses in the infrared spectrum, and the receiver detects these pulses and converts them into usable signals. Infrared communication is suitable for indoor use because it is inexpensive, simple to implement, and causes low interference between different rooms since infrared signals do not pass through walls effectively. However, infrared has a limited range and usually requires line-of-sight or near-line-of-sight communication. Its signal can be blocked by obstacles, so it is not suitable for wide-area mobility or outdoor long-distance communication.

Example:

Traditional TV remotes use infrared communication. Some older laptops and mobile devices used **IrDA** ports for short-range file transfer.

3.2.8 Satellite Communication

Satellite communication is a form of wireless communication in which signals are transmitted between **earth stations** and **satellites**. It is useful in areas where cable, fiber, or terrestrial wireless

infrastructure is difficult, expensive, or unavailable, such as remote locations, ships, aircraft, and disaster recovery sites.

In satellite communication, a ground station sends a signal upward to a satellite. This is called the **uplink**. The satellite receives the signal, amplifies or processes it, and retransmits it back to another ground station or coverage area. This return transmission is called the **downlink**. Satellites commonly use **microwave frequency bands** for uplink and downlink communication.

Unguided Media: Wireless Propagation Examples

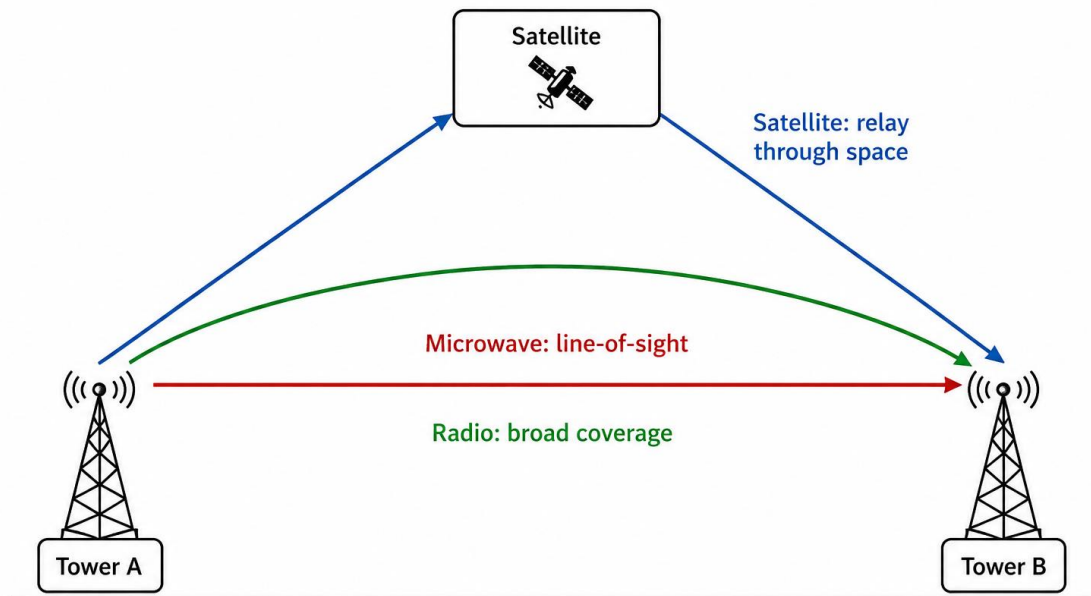


Figure 7: Wireless Propagation and Satellite Link

Satellite communication provides **wide geographic coverage** and supports broadcasting, long-distance communication, emergency connectivity, and remote-area network access. However, it may have **higher latency** because the signal must travel a long distance between Earth and space. Its performance can also be affected by weather conditions, high installation and service cost, regulatory requirements, limited capacity compared with terrestrial fiber, and the need for clear satellite visibility.

Example:

- **DTH television service**, where TV signals are sent from a broadcasting station to a satellite and then received by dish antennas at homes.
- **Internet connectivity in remote villages**, mountains, ships, and aircraft.
- **GPS navigation**, where satellites send location signals to mobile phones, vehicles, and navigation devices.

Table 4: Comparison of Unguided Transmission Media

Medium	Direction	Typical Range	Main Use	Key Limitation
Radio Wave	Mostly omnidirectional	Short to wide area depending on power and frequency	Broadcast, Wi-Fi, mobile communication	Interference and security exposure
Microwave	Directional	Point-to-point or cellular coverage	Tower links, cellular, wireless LAN backhaul	Line-of-sight and weather sensitivity
Infrared	Short-range directional or diffuse	Room-level or device-level	Remote controls, short-range device communication	Blocked by walls and obstacles
Satellite	Directional Earth-space link	Regional to global	Remote connectivity, broadcast, disaster communication	Higher latency and cost

3.3 Ethernet Cable Standards

Ethernet cable standards are physical-layer standards that define how **Ethernet data or Ethernet frames** are carried over different physical media such as **coaxial cable, twisted-pair copper cable, and optical fiber**.

Different Ethernet cable standards define the speed, cable type, signaling method, connector, and maximum distance used for communication.

For example, **10BASE5** defines Ethernet over thick coaxial cable, **100BASE-TX** defines Fast Ethernet over twisted-pair copper cable, and **1000BASE-SX** defines Gigabit Ethernet over short-distance fiber.

Table 5: Common Ethernet Naming Pattern

Part of Name	Meaning	Example
10, 100, 1000, 10G	Approximate data rate in Mbps or Gbps.	1000Base-T means 1000 Mbps or 1 Gbps Ethernet.
Base	Baseband transmission, meaning the link uses the medium for one digital signal system at a time.	10Base-T uses baseband signaling.
T / TX	Twisted-pair copper implementation.	100Base-TX uses two pairs of twisted-pair cable.
F / FX	Fiber implementation or fiber reach/wavelength class.	100Base-FX uses fiber; 1000Base-SX uses short-wavelength fiber.
5 / 2	Approximate maximum segment length in hundreds of meters for classic coaxial Ethernet.	10Base5 used thick coaxial cable; 10Base2 used thin coaxial cable.

3.3.1 Classic 10 Mbps Ethernet Cable Standards

The early generation of **Standard Ethernet** operated at 10 Mbps. It included coaxial and twisted-pair options. Older coaxial Ethernet used a shared medium and collision-based access. Later twisted-pair Ethernet moved toward hub-based and then switch-based LANs.

Table 6: Classic Ethernet Cable Standards

Standard	Speed	Medium	Practical Notes
10Base5	10 Mbps	Thick coaxial cable	Known as Thick Ethernet or

			Thicknet; historically used for backbone-like bus segments.
10Base2	10 Mbps	Thin coaxial cable	Known as Thin Ethernet or Cheapernet; easier than 10Base5 but still shared bus.
10Base-T	10 Mbps	Twisted-pair cable	Uses star wiring with hubs/switches; important step toward modern Ethernet cabling.
10Base-F	10 Mbps	Optical fiber	Used where fiber distance or EMI resistance was required.

3.3.2 Fast Ethernet Cable Standards

Fast Ethernet increased Ethernet speed from 10 Mbps to 100 Mbps. It preserved the general Ethernet frame format and made migration easier for organizations. Fast Ethernet was widely used in office networks and campus networks before Gigabit Ethernet became common.

Table 7: Fast Ethernet Cable Standards

Standard	Speed	Medium	Common Use
100Base-TX	100 Mbps	Two pairs of twisted-pair cable	Very common office LAN implementation.
100Base-FX	100 Mbps	Two optical fibers	Used for longer distance and EMI-resistant links.
100Base-T4	100 Mbps	Four pairs of voice-grade or better twisted pair	Designed to support lower-grade cable; now uncommon.

3.3.3 Gigabit Ethernet Cable Standards

Gigabit Ethernet increased the standard LAN speed to 1000 Mbps or 1 Gbps. It became common for desktop uplinks, server connections, access switches, and building distribution. Gigabit Ethernet may run over copper or fiber depending on distance and performance requirements.

Table 8: Gigabit Ethernet Cable Standards

Standard	Speed	Medium	Practical Use
1000Base-T	1 Gbps	Twisted-pair copper	Common for desktops, branch LANs, printers, IP phones, and access switch ports.
1000Base-SX	1 Gbps	Short-wavelength multimode fiber	Common for short fiber links inside buildings or data centers.
1000Base-LX	1 Gbps	Long-wavelength fiber	Used for longer fiber runs.
1000Base-CX	1 Gbps	Short copper cable	Short-distance equipment connections; less common today.

3.3.4 Twisted-Pair Cable Categories in Modern LANs

Modern Ethernet over copper commonly uses cable categories such as Cat5e, Cat6, and Cat6A. These categories are not the same as Ethernet protocols, but they strongly influence which Ethernet speeds can be reliably supported over a given distance.

Table 9: Common Twisted-Pair Cable Categories

Cable Category	Typical Ethernet Support	Maximum Distance / Practical Use	Important Note
Cat5e	1 Gbps	Up to 100 meters	Enhanced Cat5; common for basic Gigabit LANs.
Cat6	1 Gbps; 10 Gbps over shorter distance	100 meters for 1 Gbps; around 55 meters for 10 Gbps	Better than Cat5e; common in offices.
Cat6A	10 Gbps	Up to 100 meters	“A” means augmented; good for new enterprise cabling.
Cat7	10 Gbps	Up to 100 meters	ISO/IEC-recognized, but not TIA-standard; less common in normal enterprise LANs.
Cat7A	10 Gbps	Up to 100 meters	Advanced shielded category; limited practical adoption.
Cat8	25 Gbps / 40 Gbps	Up to 30 meters	Mainly for short data center or server-to-switch links.

Common misunderstanding: Students sometimes confuse cable category with Ethernet speed. Cable category describes physical cable quality. Ethernet speed is defined by the Ethernet standard and supported by network interfaces and switches. Both must match for reliable operation.

3.4 Ethernet, Fast Ethernet and Gigabit Ethernet

Ethernet is a widely used **wired Local Area Network** technology that allows computers, switches, routers, printers, servers and other devices to communicate using network cables.

In simple terms, Ethernet is the technology used when a computer is connected to a switch or router using a **LAN cable** or **RJ45 cable**. The cable is the physical path, but Ethernet is the communication technology that defines how data is transmitted over that path.

3.4.1 Standard Ethernet

Standard Ethernet is the earlier form of Ethernet technology. Its speed is **10 Mbps**, meaning it can theoretically transmit 10 million bits per second.

Standard Ethernet was important in early LANs because it allowed multiple computers to share resources such as files and printers. However, due to modern traffic demands, it is now mostly outdated for production networks.

Common Standard Ethernet Implementations

Table 2: Common Standard Ethernet Standards

Standard Name	Speed	Cable / Medium	Simple Meaning
10BASE5	10 Mbps	Thick coaxial cable	Thick Ethernet
10BASE2	10 Mbps	Thin coaxial cable	Thin Ethernet
10BASE-T	10 Mbps	Twisted-pair cable	Ethernet over LAN cable

Standard Name	Speed	Cable / Medium	Simple Meaning
10BASE-F	10 Mbps	Fiber-optic cable	Ethernet over fiber

Application of Standard Ethernet

- Small offices with very low network traffic
- Early computer labs
- Basic file sharing
- Printer sharing
- Low-speed LAN communication

3.4.2 Fast Ethernet

Fast Ethernet is an improved version of Standard Ethernet. Its speed is **100 Mbps**, which is **10 times faster** than Standard Ethernet.

Fast Ethernet was developed because normal Ethernet became insufficient for growing LAN traffic such as file sharing, internet access, email, database access and client-server applications.

Common Fast Ethernet Implementations

Table 3: Common Fast Ethernet Standards

Standard Name	Speed	Cable / Medium	Common Use
100BASE-TX	100 Mbps	Two pairs of twisted-pair cable	Common office LAN
100BASE-FX	100 Mbps	Fiber-optic cable	Longer-distance LAN links
100BASE-T4	100 Mbps	Four pairs of twisted-pair cable	Older twisted-pair implementation

Important Feature: Autonegotiation

Autonegotiation allows two connected Ethernet devices to automatically agree on the best speed and mode they both support.

Application of Fast Ethernet

- Small and medium office LANs
- Computer labs
- Printer sharing networks
- Basic internet access networks
- Older branch office networks

3.4.3 Gigabit Ethernet / Giga-Ethernet

Gigabit Ethernet, also called **Giga-Ethernet**, is a higher-speed Ethernet technology. Its speed is **1000 Mbps**, which is equal to **1 Gbps**.

- 10 times faster than Fast Ethernet
- 100 times faster than Standard Ethernet

Common Gigabit Ethernet Implementations

Table 4: Common Gigabit Ethernet Standards

Standard Name	Speed	Cable / Medium	Common Use
1000BASE-T	1 Gbps	Four twisted-pair cable pairs	Modern office LAN

Standard Name	Speed	Cable / Medium	Common Use
1000BASE-SX	1 Gbps	Short-wavelength fiber	Short-distance fiber links
1000BASE-LX	1 Gbps	Long-wavelength fiber	Longer-distance fiber links

Application of Gigabit Ethernet

- Corporate LANs
- Banking branch networks
- Data centers
- Server-to-switch connections
- Switch-to-switch uplinks
- CCTV and IP phone networks
- High-speed internet sharing
- File server and backup networks

3.4.4 Comparison of Ethernet, Fast Ethernet and Gigabit Ethernet

Table 5: Ethernet vs Fast Ethernet vs Gigabit Ethernet

Basis of Comparison	Ethernet	Fast Ethernet	Gigabit Ethernet
Speed	10 Mbps	100 Mbps	1000 Mbps / 1 Gbps
Relative Speed	Basic speed	10 times faster than Ethernet	10 times faster than Fast Ethernet
Common Standards	10BASE-T, 10BASE2, 10BASE5, 10BASE-F	100BASE-TX, 100BASE-FX, 100BASE-T4	1000BASE-T, 1000BASE-SX, 1000BASE-LX
Common Cable	Coaxial, twisted pair, fiber	Twisted pair, fiber	Twisted pair, fiber
Typical Use	Older LANs	Small office LANs	Modern LANs and data centers
Current Relevance	Mostly outdated	Still found in older networks	Very common in modern networks
Performance	Low	Medium	High
Best Example	Old office LAN	Basic branch LAN	Server, switch and high-speed LAN links

3.5 IEEE 802.11 (Wi-Fi) and IEEE 802.15.1 (Bluetooth) Standards

Wireless communication allows devices to exchange data without using physical cables. In computer networks, two important wireless standards are IEEE 802.11, commonly known as Wi-Fi, and IEEE 802.15.1, commonly associated with Bluetooth.

Both Wi-Fi and Bluetooth use wireless signals, but they are designed for different purposes. Wi-Fi is mainly used for wireless LAN and Internet access. Bluetooth is mainly used for short-range personal device communication.

The IEEE 802 family defines standards for local area networks (LANs) and metropolitan area networks (MANs). A standard is a set of technical rules that devices must follow so that they can communicate correctly.

These standards are important because network devices are made by many different companies. A laptop from one company can connect to a router or access point from another company because both follow the same standard.

Table 2: Common IEEE 802 Standards

IEEE Standard	Technology	Main Purpose
IEEE 802.3	Ethernet	Wired LAN communication
IEEE 802.11	Wi-Fi	Wireless LAN communication
IEEE 802.15.1	Bluetooth	Wireless personal area communication

Example: A student may use a laptop of one brand, a Wi-Fi access point of another brand, and a mobile phone of another brand. These devices can still connect because they follow common wireless standards.

Analogy: Standards are like traffic rules. If everyone follows the same rules, vehicles can move safely on the same road. Similarly, if devices follow the same network standard, they can communicate properly.

3.5.1 IEEE 802.11 Wi-Fi

IEEE 802.11 is a family of standards for wireless local area networks (WLANs). It is commonly called Wi-Fi. Wi-Fi allows devices such as laptops, mobile phones, tablets, printers, and IoT devices to communicate without using network cables.

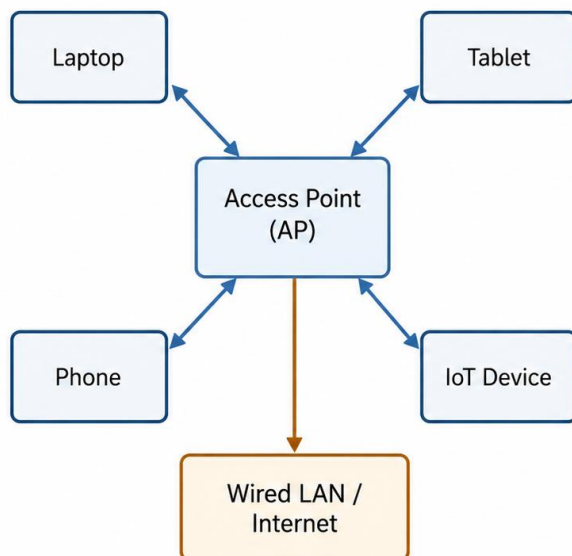
Wi-Fi uses radio waves to carry data through the air. These radio waves move between the wireless device and an access point or, in some cases, directly between devices.

Purpose: The main purpose of Wi-Fi is to provide wireless access to a LAN or the Internet. It gives users mobility, meaning they can move within the coverage area without disconnecting from the network.

Example: In a college, students connect their laptops and mobile phones to campus Wi-Fi to access online classes, learning materials, email, and the Internet.

IEEE 802.11 Wi-Fi: Infrastructure and Ad Hoc Mode

Infrastructure Mode



Ad Hoc Mode

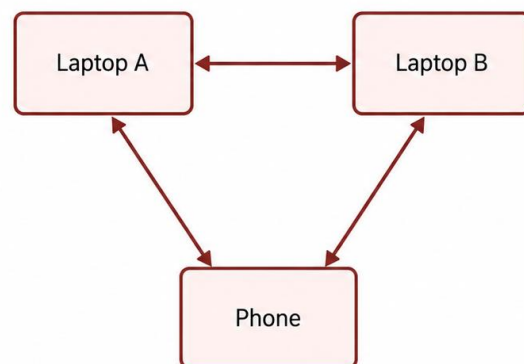


Figure 10: IEEE 802.11 Wi-Fi Infrastructure Architecture**Wi-Fi Architecture**

Wi-Fi commonly works in two modes: infrastructure mode and ad hoc mode.

Infrastructure mode: In infrastructure mode, wireless devices connect through an access point (AP). The AP acts as the central device that connects wireless users to a wired LAN or the Internet. Most home, office, campus, hotel, and airport Wi-Fi networks use infrastructure mode.

Ad hoc mode: In ad hoc mode, devices communicate directly with each other without using an access point. This is useful for temporary peer-to-peer communication, but it is less common in enterprise and campus networks.

Table 11: Selected IEEE 802.11 Standards

Wi-Fi Generation	IEEE Standard	Maximum Theoretical Speed	Frequency Band Used
Wi-Fi 1	802.11b	11 Mbps	2.4 GHz
Wi-Fi 2	802.11a	54 Mbps	5 GHz
Wi-Fi 3	802.11g	54 Mbps	2.4 GHz
Wi-Fi 4	802.11n	600 Mbps	2.4 GHz & 5 GHz
Wi-Fi 5	802.11ac	6.9 Gbps	5 GHz
Wi-Fi 6	802.11ax	9.6 Gbps	2.4 GHz & 5 GHz
Wi-Fi 6E	802.11ax (Ext)	9.6 Gbps	2.4 GHz, 5 GHz & 6 GHz
Wi-Fi 7	802.11be	46 Gbps	2.4 GHz, 5 GHz & 6 GHz

3.5.2 IEEE 802.15.1 Bluetooth

Bluetooth is a short-range wireless technology designed to connect personal devices. Unlike Wi-Fi, which is intended for local area networking, Bluetooth is mainly designed for wireless personal area networks. It connects devices such as phones, headsets, keyboards, mice, printers, sensors, and wearable devices.

Bluetooth Piconet

A Bluetooth network is called a piconet. A piconet has one primary device and up to seven active secondary devices. The secondary devices synchronize their timing and communication behavior with the primary device.

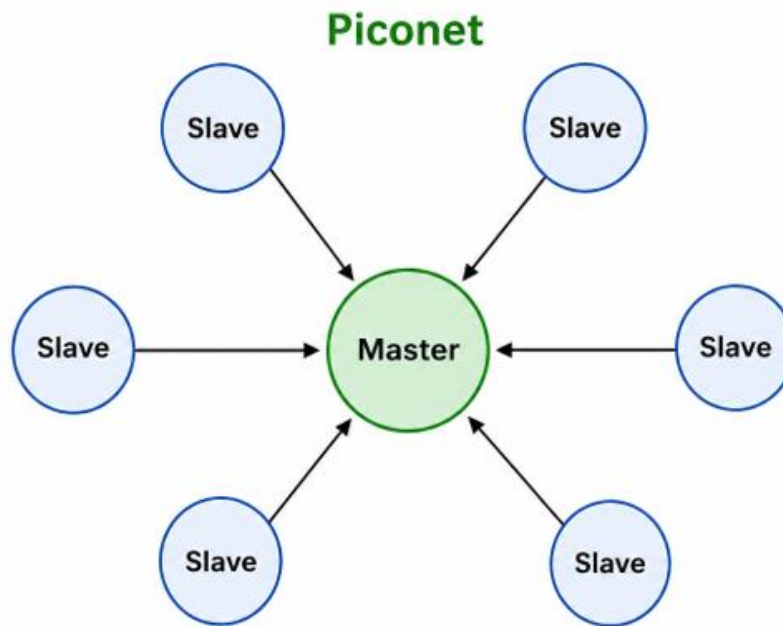


Figure: Piconet

Example: A mobile phone connected to Bluetooth earphones, a smartwatch, and a wireless speaker can be understood as a small Bluetooth personal network.

Bluetooth Scatternet

A scatternet is formed when multiple piconets are connected. This can happen when one device participates in more than one piconet. A device may act as a secondary in one piconet and as a primary in another piconet.

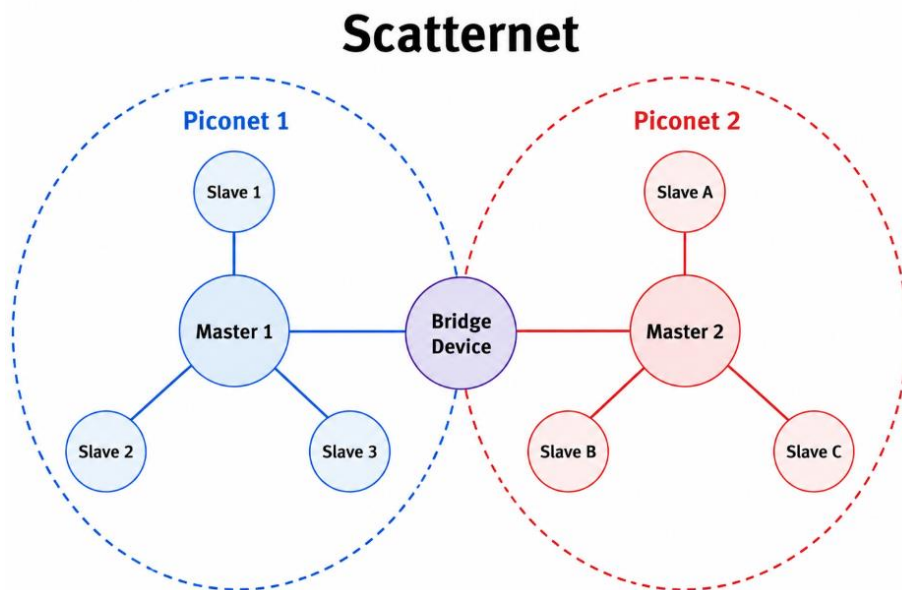


Figure: Scatternet

Simple idea: A piconet is one small Bluetooth group. A scatternet is formed when two or more small Bluetooth groups are connected through a shared device.

Table 12: Wi-Fi vs Bluetooth Comparison

Basis	Wi-Fi (IEEE 802.11)	Bluetooth (IEEE 802.15.1)
Network type	Wireless LAN	Wireless personal area network
Main purpose	Internet/LAN access and high-speed local connectivity	Short-range device-to-device connectivity
Typical range	Room, building, campus depending on AP design	Short range around a person, desk, or room
Power use	Generally higher than Bluetooth	Designed for low power operation
Architecture	Access point infrastructure or ad hoc mode	Piconet and scatternet
Examples	Campus Wi-Fi, corporate WLAN, home router	Wireless mouse, headset, smartwatch, sensor

Common misunderstanding: Wi-Fi and Bluetooth are not just two names for wireless communication. Wi-Fi is mainly for network access; Bluetooth is mainly for short-range personal device communication.

Applications

Wi-fi

- Home Wi-Fi Networks: Connecting phones, laptops, and smart TVs to high-speed internet.
- Public Hotspots: Accessing free or guest internet in cafes, airports, and hotels.
- Smart Home Ecosystems: Linking bandwidth-heavy devices like security cameras and smart displays.
- Wireless File Transfer: Sharing files directly between devices over the same network.
- Enterprise Networking: Providing wireless internet and secure local intranet for large office buildings and campuses.

Bluetooth

- Wireless Audio: Connecting earbuds, headphones, and portable speakers.
- Device Peripherals: Linking wireless mice, keyboards, and gaming controllers.
- Wearable Tech: Syncing smartwatches and fitness trackers to smartphones.
- Proximity Tracking: Finding lost keys or luggage using item trackers (like AirTags).

3.6 Signals

A **signal** is the physical form used to carry **data** from one device to another through a communication medium. In networking, **data** cannot travel by itself. It must be converted into a **signal** before it can move through a cable, fiber, or wireless medium.

For example:

- In copper cable, data travels as electrical signals.
- In optical fiber, data travels as light signals.
- In Wi-Fi or mobile networks, data travels as electromagnetic radio waves.

In simple words, **data** is the information, while **signal** is the physical carrier of that information. Data may be a message, file, image, video, transaction record, or email. A **signal** is the physical form that carries that **data** from sender to receiver.

Example: When a user opens a banking website, the login request is data. During transmission, that data may travel as electrical signals inside a computer, light pulses through fiber, and radio waves through Wi-Fi.

3.6.1 Analog Signals and Digital Signals

Signals can also be **analog** or **digital**.

An **analog signal** varies continuously with time. It has many possible values within a range. A sine wave is a common example of an **analog signal**.

A **digital signal** changes in discrete levels. It commonly represents binary values such as 0 and 1. Instead of changing smoothly like an analog signal, a **digital signal** changes from one level to another.

Example: A computer stores an image as digital data. When the image is sent through a Wi-Fi network, the data is carried using radio-frequency signals.

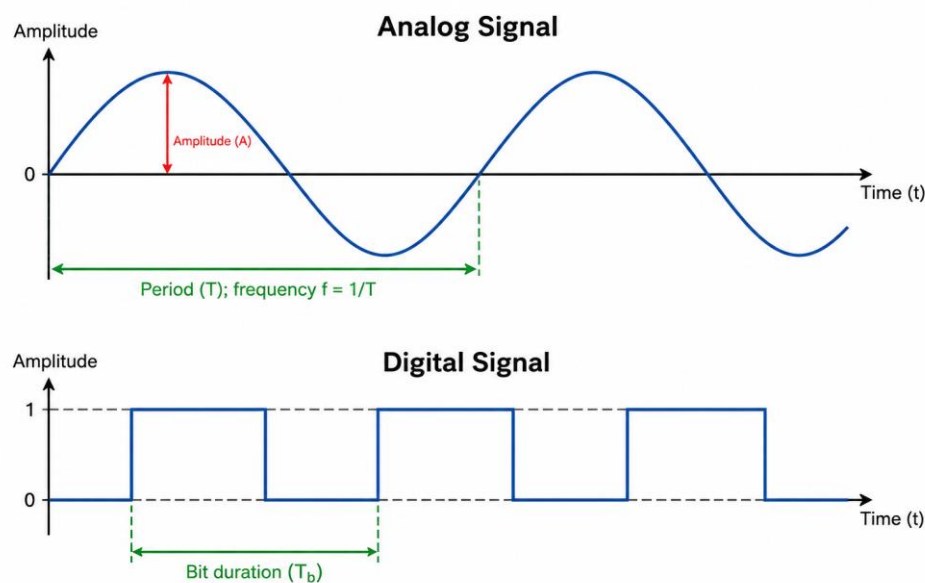


Figure 12: Analog and Digital Signal

3.6.2 Why Signals Are Important in Networking

Signals are important because every network communication depends on successful signal transmission. If the signal becomes weak, distorted, or affected by noise, the receiver may not correctly understand the transmitted bits.

Studying signals helps us understand:

- why cable quality matters;
- why distance affects communication;
- why fiber supports long-distance high-speed communication;

- why Wi-Fi speed decreases with weak signal;
- why interference causes errors;
- why higher speed needs better bandwidth and signal quality.

Example: A damaged Ethernet cable may still show a link light, but poor signal quality may cause packet loss, slow speed, or frequent disconnection.

Analogy: A signal is like someone speaking in a room. If the person speaks clearly and loudly, the listener understands the message. If the voice is weak, distorted, or mixed with noise, the listener may misunderstand. Network signals behave in a similar way.

3.7 Transmission Impairment: Attenuation, Distortion and Noise

Transmission impairment means that the received signal is not exactly the same as the transmitted signal. In any real network, signals face loss, delay differences, interference, and unwanted energy. Impairment is important because data communication depends on the receiver correctly interpreting the signal.

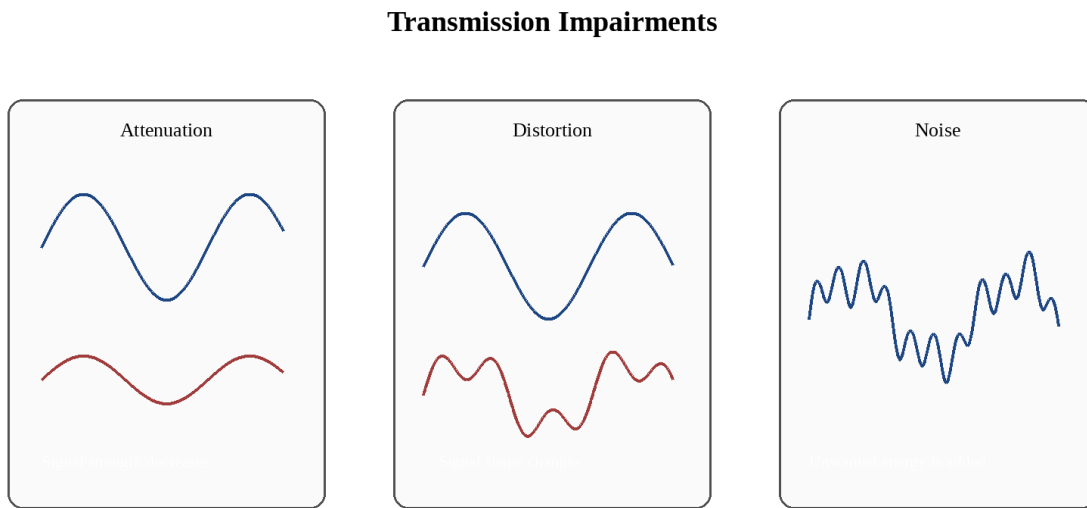


Figure 14: Major Types of Transmission Impairment

3.7.1 Attenuation

Attenuation is the loss of signal strength as the signal travels through a medium. In copper cables, electrical energy is gradually lost as heat. In fiber, light power also reduces over distance. In wireless, signal strength spreads and weakens as distance increases.

Table 2: Attenuation in Different Media

Medium	How Attenuation Happens	Simple Example
Copper cable	Electrical energy is partly lost as heat due to resistance.	Long or poor-quality UTP cable may reduce Ethernet signal quality.

Optical fiber	Light power gradually reduces as it travels through the fiber.	Long fiber links require proper optical power budget.
Wireless medium	Radio signal spreads out and weakens with distance and obstacles.	Wi-Fi signal becomes weak far from the access point.

Formula:

$$dB = 10 \log_{10}(P_2 / P_1)$$

Where P1 is the input power and P2 is the output power. If the dB value is negative, the signal has lost power. If the dB value is positive, the signal has gained power.

Solved Example:

A signal enters a cable with power P1 = 10 mW and leaves with power P2 = 5 mW.

1. $dB = 10 \log_{10}(P_2 / P_1)$
2. $dB = 10 \log_{10}(5 / 10)$
3. $dB = 10 \log_{10}(0.5)$
4. $dB \text{ approximately } = -3.01 \text{ dB}$

The negative value shows attenuation, meaning the signal power decreased.

Practice Question:

A signal enters a cable at 20 mW and leaves at 2 mW. Calculate the attenuation in dB.

3.7.2 Distortion

Distortion means that the signal changes its shape or form during transmission. A signal may still be strong, but if its shape has changed, the receiver may not interpret it correctly. Distortion is especially important in composite signals, which are made of multiple frequency components.

Different frequency components may travel through a medium with different speeds or different delays. Because of this, they may arrive at the receiver with changed timing or phase. The final received signal shape becomes different from the original signal shape.

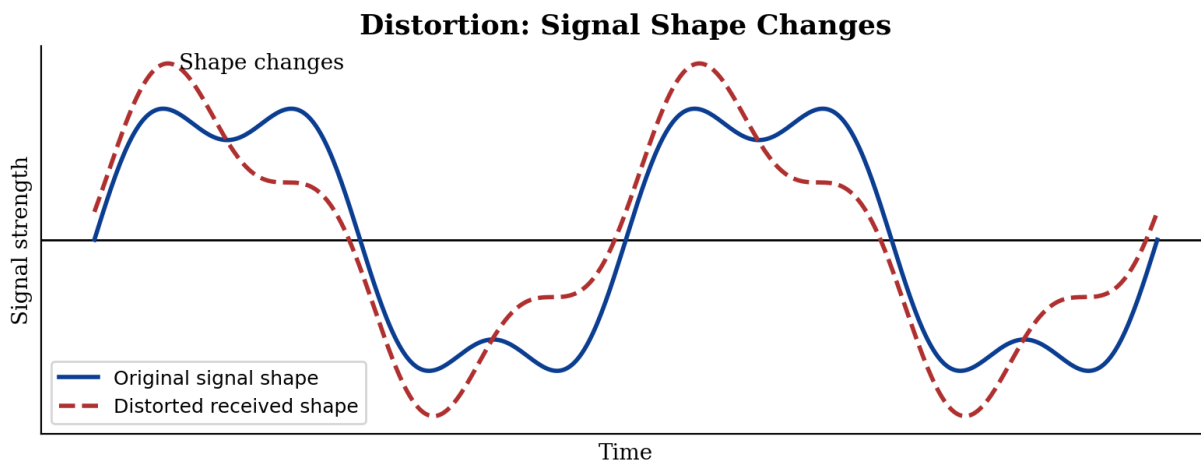


Figure 3: Distortion - Signal Shape Changes

Why distortion happens

Distortion may occur because:

- different parts of the signal travel at different speeds;
- the medium does not treat all frequencies equally;
- bandwidth of the medium is not enough;
- poor cable quality affects high-speed signals.

Example:

In a high-speed copper network, a poor-quality cable may distort the signal transitions. The receiver may see unclear bit boundaries and may misread a 0 as a 1 or a 1 as a 0.

Common misunderstanding:

Distortion is not the same as attenuation. In attenuation, the signal becomes weaker. In distortion, the signal shape changes. A distorted signal may still be strong, but its form may be incorrect.

3.7.3 Noise

Noise is unwanted energy that is added to the original signal during transmission. Noise may come from electronic components, nearby electrical equipment, other cables, lightning, radio interference, or sudden electromagnetic disturbances.

Noise is dangerous because it can hide or change the original signal. In digital communication, noise may cause the receiver to interpret a 0 as a 1 or a 1 as a 0.

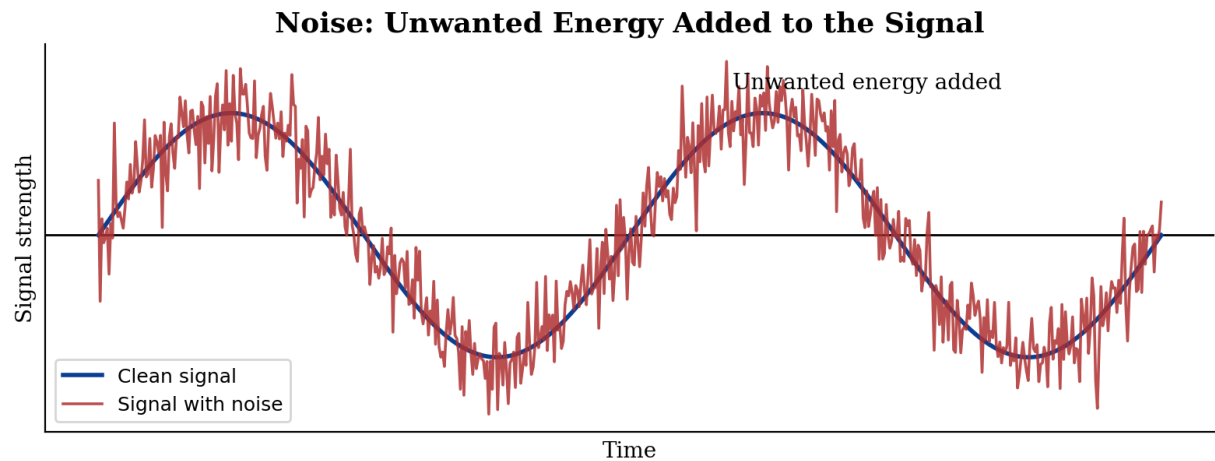


Figure 4: Noise - Unwanted Energy Added to the Signal

Table 4: Common Types of Noise

Noise Type	Meaning	Example
Thermal noise	Random motion of electrons creates unwanted signal energy.	Background electrical noise in circuits and copper wires.
Induced noise	Noise created by external electromagnetic sources.	Motors, power lines, fluorescent lights, or radio transmitters.
Crosstalk	Signal from one wire affects another nearby wire.	Poorly installed twisted-pair cables in a cable bundle.
Impulse noise	Sudden high-energy disturbance for a short time.	Lightning, switching equipment, or power spikes.

Signal-to-Noise Ratio (SNR)

Signal-to-Noise Ratio (SNR) compares the strength of the useful signal with the strength of noise. A higher SNR means the useful signal is stronger than the noise, so communication quality is usually better.

Formula:

$$\text{SNR} = \text{Signal Power} / \text{Noise Power}$$

$$\text{SNR}_{\text{dB}} = 10 \log_{10}(\text{SNR})$$

Solved Example:

If signal power is 100 times the noise power, then $\text{SNR} = 100$.

$$5. \text{SNR}_{\text{dB}} = 10 \log_{10}(100)$$

$$6. \text{SNR}_{\text{dB}} = 10 \times 2$$

$$7. \text{SNR}_{\text{dB}} = 20 \text{ dB}$$

This means the signal is much stronger than the noise.

3.7.4 Reducing Transmission Impairments

- Use high-quality cables and connectors appropriate for the required speed.
- Avoid running data cables parallel to high-voltage electrical cables.
- Use shielding or optical fiber in electrically noisy environments.
- Respect maximum cable length and use repeaters, switches, or fiber uplinks where required.
- Design Wi-Fi coverage using proper channel planning and access point placement.
- Monitor error rates, signal strength, interface drops, and retransmissions.
- Use error detection, retransmission, and correction methods in higher layers where appropriate.

Common misunderstanding: A strong signal is not always a good signal. A signal may have enough strength but still be distorted or noisy. Quality depends on strength, shape, timing, and noise level.

3.8 Multiplexing: TDM, FDM and WDM

3.8.1 Meaning and Importance of Multiplexing

Multiplexing is a technique that allows multiple signals or data streams to share one physical link. A **multiplexer (MUX)** combines several input lines into one shared link. A **demultiplexer (DEMUX)** separates the combined signal back into individual outputs at the receiving end.



Figure 15: Multiplexing Using MUX and DEMUX

Multiplexing is important because communication links are expensive. Instead of installing separate physical links for every conversation or data stream, organizations can share the capacity of a single link among many users or services.

Analogy: Multiplexing is like a highway used by many vehicles. Instead of building a separate road for each traveler, multiple vehicles share the same road by using different lanes, different times, or different rules.

3.8.2 Why Multiplexing is Important in Communication Systems

Multiplexing is important because it improves bandwidth utilization. Bandwidth utilization means using the available communication capacity effectively. In real networks, communication links are valuable resources, so they should not remain unused or underused.

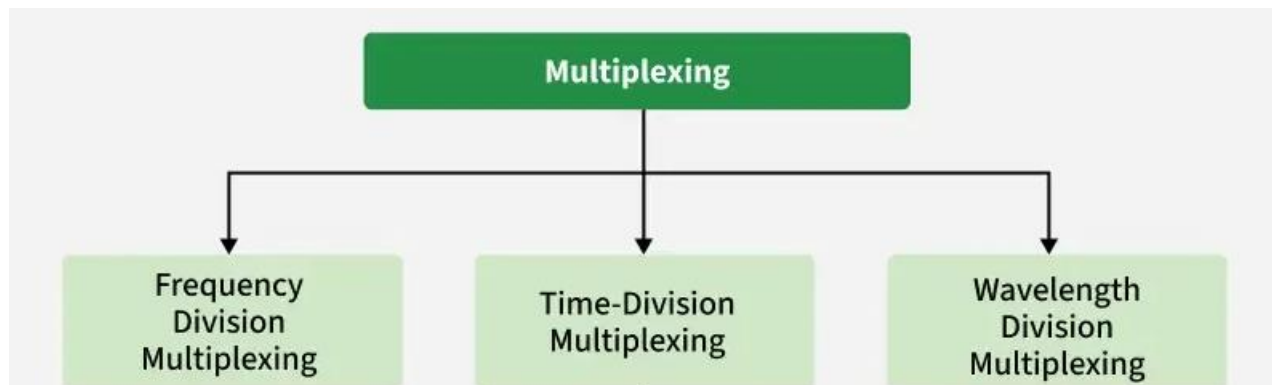


Figure: Multiplexing

- It reduces the need for many separate physical links.
- It lowers communication infrastructure cost.
- It improves use of available bandwidth.
- It allows many users or services to communicate over the same link.
- It supports telephone networks, cable TV, radio systems, fiber backbone networks, and data center links.
- It helps organizations carry voice, video, and data over shared communication links.

Example: In a bank, multiple branches may communicate with the data center through shared telecom infrastructure. Multiplexing allows several communication streams such as ATM traffic, branch application traffic, email, and monitoring data to use common backbone links efficiently.

3.8.2 Frequency-Division Multiplexing (FDM)

Frequency-Division Multiplexing (FDM) is a multiplexing technique in which the available bandwidth of a communication link is divided into several separate frequency bands. Each signal is assigned its own frequency range, so many signals can travel through the same physical medium at the same time. In simple words, FDM allows multiple users or services to share one link by giving each one a different frequency channel.

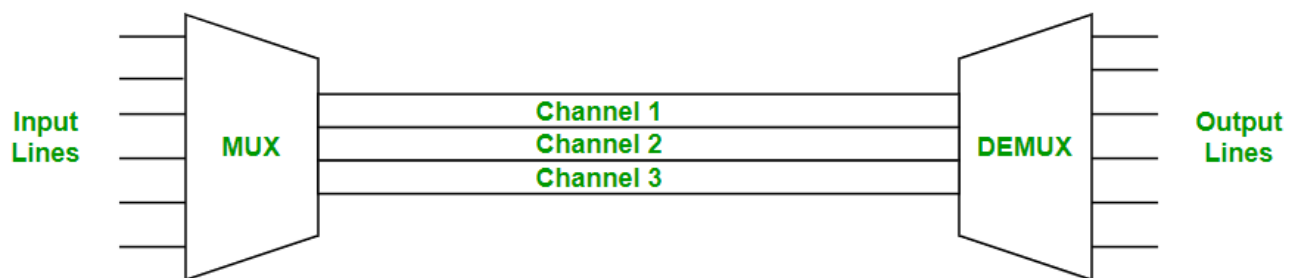


Figure: FDM

FDM is mainly used in analog communication systems. In this method, each source signal is placed on a different carrier frequency. These modulated signals are then combined into one composite signal and transmitted together through the shared link. At the receiving side, the demultiplexer separates the signals according to their frequency ranges and sends each signal to the correct receiver. To prevent overlap between nearby channels, small unused frequency gaps called **guard bands** are used.

Example:

Radio broadcasting is a common example of FDM. Different radio stations use different frequencies, such as 98.3 MHz, 100.0 MHz, or 102.4 MHz. All these signals are present in the air at the same time, but a radio receiver selects only the frequency of the station the listener wants to hear.

Applications:

FDM is used in radio broadcasting, television broadcasting, cable television, older telephone carrier systems, and some broadband communication systems.

3.8.3 Time-Division Multiplexing (TDM)

Time-Division Multiplexing (TDM) is a multiplexing technique in which multiple signals share the same communication link by using different time slots. Instead of dividing the link by frequency, TDM divides the use of the link by time. Each source gets a short time period to transmit data, and then the next source gets its turn.

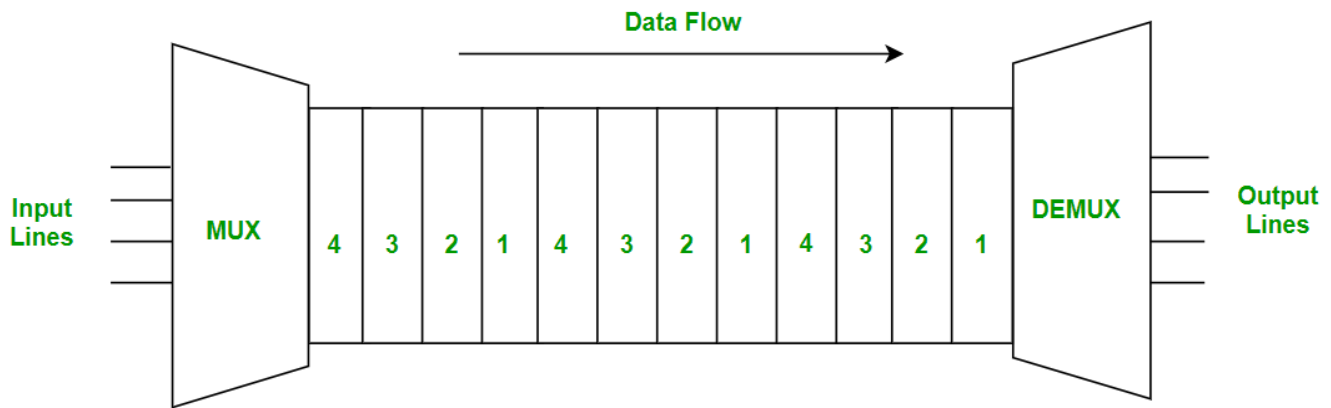


Figure: TDM

In TDM, the full bandwidth of the channel is available to one source during its assigned time slot. The time slots are very small and repeat continuously, so communication appears to happen at the same time for all users. At the receiving side, the demultiplexer separates the data based on the order and timing of the slots. TDM requires proper timing and synchronization between sender and receiver.

Example:

In a digital telephone system, many voice calls can share one high-speed trunk line. Each call is given a small time slot repeatedly, so many calls can be carried over the same physical link.

Applications:

TDM is used in digital telephony, digital communication links, leased lines, backbone networks, and systems where multiple digital data streams need to share one link.

3.8.4 Wavelength-Division Multiplexing (WDM)

Wavelength-Division Multiplexing (WDM) is a multiplexing technique used in optical fiber communication. It combines multiple light signals of different wavelengths and sends them through a single fiber-optic cable. In simple words, WDM allows different “colors” of light to carry different data streams through the same fiber at the same time.

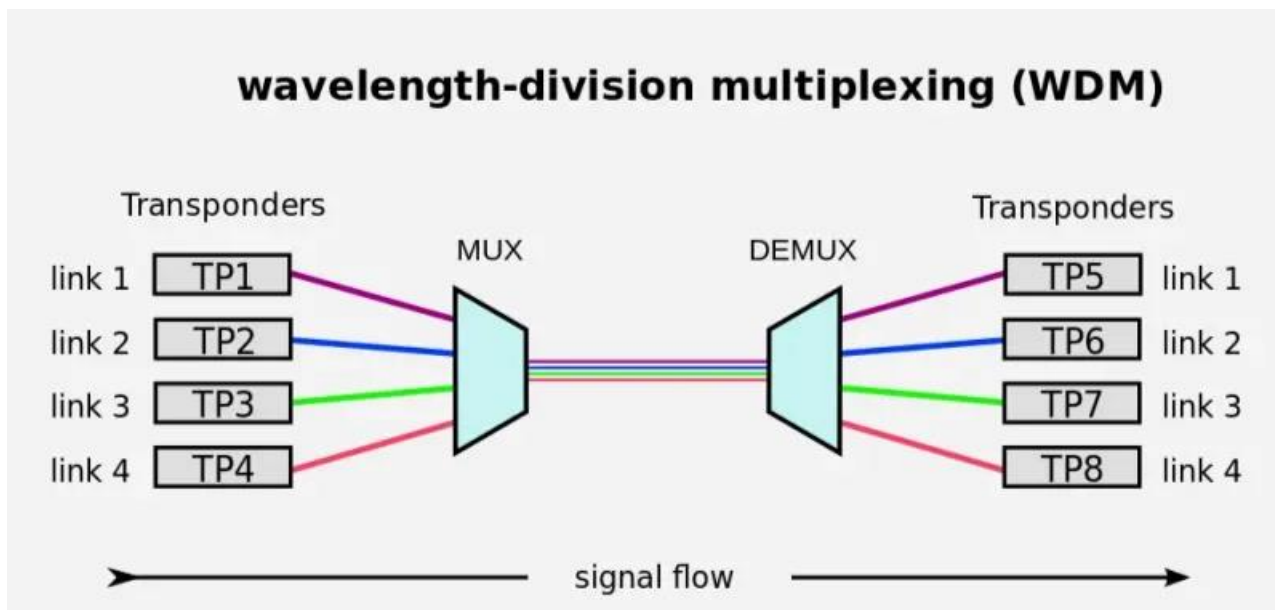


Figure: WDM

WDM is similar to FDM, but instead of using different electrical frequency bands, it uses different wavelengths of light. At the sending side, several optical signals are combined by a WDM multiplexer and transmitted through one fiber. At the receiving side, a WDM demultiplexer separates the different wavelengths and sends each signal to the correct output. WDM greatly increases the capacity of optical fiber links without requiring separate fiber cables for every signal.

There are two important forms of WDM. **Coarse Wavelength-Division Multiplexing (CWDM)** uses fewer wavelengths with wider spacing between channels and is cheaper. **Dense Wavelength-Division Multiplexing (DWDM)** uses many wavelengths placed closely together, allowing very high-capacity fiber communication.

Example:

A telecom company may use WDM to send Internet traffic, voice traffic, and data center traffic over the same fiber cable using different wavelengths of light.

Applications:

WDM is used in ISP backbone networks, telecom carrier networks, long-distance fiber communication, data center interconnection, and high-capacity enterprise WAN links.

Table 17: FDM vs TDM vs WDM

Basis	FDM	TDM	WDM
Full Form	Frequency-Division Multiplexing	Time-Division Multiplexing	Wavelength-Division Multiplexing
Sharing Basis	Frequency bands	Time slots	Light wavelengths
Main Medium	Radio, coaxial, wireless links	Digital links	Optical fiber
Transmission Style	Signals transmit at the same time using different frequencies	Signals take turns using different time slots	Multiple light wavelengths transmit at the same time

Common Example	Radio stations using different frequencies	Digital telephone trunk	Multiple light wavelengths in one fiber
Main Benefit	Supports simultaneous transmission	Efficient sharing of digital links	Very high fiber capacity
Limitation	Needs guard bands and frequency planning	Needs timing synchronization	Requires optical equipment

Common misunderstanding: Multiplexing does not mean data is mixed randomly. The signals are combined using strict rules so that the receiver can separate them correctly.

Key Takeaways

- **FDM** divides the link by frequency.
- **TDM** divides the link by time.
- **WDM** divides optical fiber by wavelength of light.
- FDM is common in radio and television broadcasting.
- TDM is common in digital telephony and digital communication.
- WDM is common in fiber backbone and telecom networks.
- Multiplexing improves bandwidth utilization and reduces the need for separate physical links.

3.9 Switching: Circuit Switching, Message Switching, Packet Switching and Virtual Circuit Switching

Switching is the process of moving data from source to destination through intermediate devices. In large networks, a sender and receiver are usually not directly connected by a single cable. Instead, data passes through switches, routers, or other network nodes. Switching determines how the path is created and how data is forwarded.

3.9.1 Circuit Switching

Circuit switching is a technique in which a dedicated communication path is established between sender and receiver before data transmission starts.

In this method, the path remains reserved until the communication ends. It is commonly associated with traditional telephone networks.

Circuit switching has three phases: connection setup, data transfer, and connection termination. It provides a fixed path and predictable communication, but it may waste resources if the connection is idle.

Example:

A traditional landline telephone call.

3.9.2 Message Switching

Message switching is a technique in which the complete message is sent from one node to another. Each intermediate node stores the full message first and then forwards it to the next node.

This is also called a **store-and-forward** method.

Message switching does not require a dedicated path. However, it can cause delay because the whole message must be stored before forwarding. It is not suitable for real-time communication.

Example:

Email delivery is similar to message switching because an email may be stored on a server before being forwarded.

3.9.3 Packet Switching

Packet switching is a technique in which data is divided into small parts called **packets** before transmission.

Each packet contains data and address information. Packets are sent through the network and reassembled at the destination.

Packet switching uses network resources efficiently because links are shared by many users. It is suitable for data communication such as web browsing, email, file transfer, and online applications. However, packets may experience delay, loss, or arrive out of order.

Example:

Internet communication uses packet switching.

3.9.4 Virtual Circuit Switching

Virtual circuit switching is a packet switching technique in which a logical path is established before packets are transmitted.

All packets follow the same logical path, but the path is not physically dedicated like circuit switching.

Virtual circuit switching combines features of circuit switching and packet switching. It provides better control than normal packet switching because packets follow a pre-decided route. However, it requires a setup phase before data transfer.

Example:

Frame Relay and ATM networks use virtual circuit concepts.

3.10 Network Performance: Bandwidth, Throughput, Latency, Bandwidth-Delay Product and Jitter

Network performance means how well a network transfers data from one device to another. A good network should provide sufficient speed, low delay, stable communication, and reliable data delivery. The main performance metrics are **bandwidth**, **throughput**, **latency**, **bandwidth-delay product**, and **jitter**.

3.10.1 Bandwidth

Bandwidth is the maximum capacity of a communication link. It tells how much data a network link can carry in one second.

In computer networks, bandwidth is usually measured in:

- **bps** = bits per second
- **Kbps** = kilobits per second
- **Mbps** = megabits per second
- **Gbps** = gigabits per second

Example:

If an Internet connection has a bandwidth of **100 Mbps**, it means the link can theoretically carry up to **100 million bits per second**.

Bandwidth is like the width of a road. A wider road can carry more vehicles. Similarly, higher bandwidth can carry more data.

3.10.2 Throughput

Throughput is the actual amount of data successfully transferred through the network per second.

Bandwidth is the maximum possible capacity, but throughput is the real achieved performance.

Example:

A user may have a **100 Mbps** Internet connection, but during actual download, the user may get only **70 Mbps**. Here:

Bandwidth = 100 Mbps

Throughput = 70 Mbps

Throughput may be lower because of:

- network congestion
- weak Wi-Fi signal
- server limitation
- packet loss
- poor cable quality
- firewall or router processing delay
- number of users sharing the network

Bandwidth is the promised road capacity, while throughput is the actual traffic successfully moving on the road.

3.10.4 Latency

Latency is the time taken for data to travel from source to destination.

It is usually measured in:

milliseconds (ms)

Low latency means data reaches quickly. High latency means data takes longer to reach the destination.

Example:

If you send a ping to a server and get a reply in **30 ms**, the latency is about **30 milliseconds**.

Latency is important for:

- video calls
- online gaming
- remote desktop
- online banking response time
- VoIP calls
- real-time applications

Latency is like travel time. Even if the road is wide, it still takes time to travel from one city to another.

3.10.5 Bandwidth-Delay Product

Bandwidth-Delay Product (BDP) shows the amount of data that can be “in transit” in a network at one time.

It is calculated as:

Bandwidth-Delay Product = Bandwidth × Delay

Where:

- **Bandwidth** = capacity of the link
- **Delay** = latency or round-trip delay
- **BDP** = amount of data that can fill the link

Example:

If a link has:

Bandwidth = 10 Mbps

Delay = 0.1 second

Then:

$BDP = 10 \text{ Mbps} \times 0.1 \text{ second}$

$BDP = 1 \text{ Megabit}$

This means about **1 Megabit of data** can be present in the network path at one time.

Importance:

BDP is useful in high-speed and long-distance networks. It helps understand how much data should be sent before waiting for acknowledgment.

BDP is like the number of vehicles that can be on a road between two cities at the same time.

3.10.6 Jitter

Jitter is the variation in packet delay. If packets are expected every 20 ms but sometimes arrive after 20 ms, sometimes after 60 ms, and sometimes after 10 ms, the delay is uneven. Jitter affects real-time applications such as voice, video conferencing, and live streaming.

Example: In a WhatsApp or Teams call, high latency means the other person hears your voice late. High jitter means your voice may sound uneven, broken, robotic, or unstable because packets arrive at inconsistent intervals.

Table 20: Network Performance Metrics Summary

Metric	Measures	Unit	Practical Meaning
Bandwidth	Theoretical capacity of link/channel	bps	Maximum possible carrying capacity
Throughput	Actual successful useful data delivery rate	bps, Mbps, Gbps	Real achieved speed
Latency	Time delay from source to destination	ms, s	How long data takes to arrive
Bandwidth-delay product	Data that can be in transit	bits or bytes	Capacity of the network pipe
Jitter	Variation in delay	ms	Uneven arrival of packets

Common misunderstanding: Bandwidth and throughput are not the same. Bandwidth is the maximum capacity; throughput is what you actually get. Similarly, latency and jitter are not the same. Latency is delay; jitter is variation in delay.